



```
1  #include <stdio.h>
2
3  int fibonacci(int n)
4  {
5      if (n == 0)
6          return 0;
7      else if (n == 1)
8          return 1;
9      return fibonacci(n - 1) + fibonacci(n - 2);
10 }
11 int main()
12 {
13     int n;
14
15     printf("Enter the value of n: ");
16     scanf("%d", &n);
17
18     printf("The %dth Fibonacci number is: %d\n", n, fibonacci(n));
19 }
```

Enter the value of n: 5

The 5th Fibonacci number is: 5

=== Code Execution Successful ===

n^{th} Fibonacci number using Recursion.

Logical steps:

1. Start
2. Read integer n
3. call recursive function $\text{fibonacci}(n)$
4. If $n == 0$, return 0
5. If $n == 1$, return 1
6. Return $\text{fibonacci}(n-1) + \text{fibonacci}(n-2)$
7. ^{Print} Result
8. Stop.

Tree Method

